

Q1. Create a Pandas DataFrame using a dictionary with at least 3 columns and 5 rows.

```
import pandas as pd
df=pd.DataFrame({
    'Roll_Number':[1,2,3,4],
    'Name':['Kanishka','Sahishnu','Chandran','Divya'],
    'Marks':[95,96,97,98]
})
print(df)
```

Q2. #Write a Python program to read a CSV file using Pandas.

```
import pandas as pd
df=pd.DataFrame({
    'Emp_id':[101,102,103],
    'Name':['Kanishka','Sahishnu','Divya'],
    'Dept':['IT','Hr','Asst.Professor'],
    'Salary':[80000,90000,100000]
})
df.to_csv('employee.csv',index=False)
print("CSV file created sucessfully")
data=pd.read_csv('employee.csv')
print(data)
```

Q3.Read an Excel file named students.xlsx and display only the first 5 records.

```
import pandas as pd
df=pd.DataFrame({
    'Roll_no':[1,2,3,4,5,6,7,8,9,10],
    'Name':['Abhi','Raju','Krish','Jeel','Prince','Karma','Jainam','Fenil','Karan','Manthan'],
    'Marks':[91,92,93,94,95,96,97,98,99,100]
})

df.to_excel('students1.xlsx',index=False)
```

```
print("Excel file is created successfully")
data=pd.read_excel('students1.xlsx')
#print(data)
#print(df.head())
#print(df.sample(3))
print(df.nlargest(3,'Marks'))
print(df.nsmallest(3,'Marks'))
```

Q4. Display the number of rows and columns in a DataFrame.

```
import pandas as pd
df=pd.DataFrame({
    'Roll_number':[1,2,3],
    'Name':['Divya','Kanishka','Sahishnu']
})
rows,columns=df.shape
print("Number of rows -",rows)
print("Number of coloums",columns)
```

Q5. Print the data types of each column in a DataFrame.

```
import pandas as pd
df=pd.DataFrame({
    'Roll_number':[1,2,3],
    'Name':['Divya','Kanishka','Sahishnu'],
    'Marks':[100,98,89]
})
print(df.dtypes)
```

Q6. Write code to display the last 3 rows of a DataFrame.

```
import pandas as pd
df=pd.DataFrame({
    'Roll_no':[1,2,3,4,5,6,7,8,9,10],
    'Name':['Abi','Raju','Krish','Jeel','Prince','Karma','Jainam','Fenil','Karam','Manthan'],
```

```
    'Marks':[91,92,93,94,95,96,97,98,99,100]})  
print(df.tail(3))
```

Q7. Select and display only one column from a DataFrame.

```
import pandas as pd  
df=pd.DataFrame({  
    'Roll_number':[1,2,3],  
    'Name':['Divya','Kanishka','Sahishnu'],  
    'Marks':[100,98,89]  
})  
print(df['Name'])
```

Q8. Check whether a DataFrame contains missing values.

```
import pandas as pd  
df=pd.DataFrame({  
    'Roll_number':[None,2,3],  
    'Name':['Divya',None,'Sahishnu'],  
    'Marks':[100,98,None]  
})  
print(df.isnull())
```

Q9. Replace all missing values with 0.

```
import pandas as pd  
df=pd.DataFrame({  
    'Roll_number':[None,2,3],  
    'Name':['Divya',None,'Sahishnu'],  
    'Marks':[100,98,None]  
})  
print("Original Data frame")  
print(df)  
df_filled= df.fillna(0)
```

```
print("\nDataFrame after replacing missing values with 0:")
print(df_filled)
```

Q10. Write a command to sort a DataFrame based on a column in ascending order.

```
import pandas as pd
# Sample DataFrame
df = pd.DataFrame({
    'Roll_No': [3, 1, 5, 2, 4],
    'Name': ['Amit', 'Riya', 'Neha', 'Karan', 'Priya'],
    'Marks': [78, 85, 92, 69, 88]
})
# Sort DataFrame based on 'Marks' in ascending order
df_sorted = df.sort_values(by='Marks')
print(df_sorted)
```

Q11. Write a program that:

- 1. Reads student attendance data from an Excel file.**
- 2. Calculates the overall attendance percentage for each student based on multiple subjects.**
- 3. Determines exam eligibility (Eligible if percentage ≥ 71 , otherwise Not Eligible).**
- 4. Displays the final ERP attendance report.**
- 5. Exports the report to a new Excel file for record-keeping.**

Code: -

```
import pandas as pd

df=pd.read_excel('attendance.xlsx')

subject=['DS','PHP','SWT','ADW','ADJ']

df['Overall_percentage']=df[subject].mean(axis=1).round(2)

df['Exam_Status']= df['Overall_percentage'].apply(
    lambda x:'Eligible' if x>=71 else 'Not Eligible' )

erp_report= df[['Roll_No','Name']+subject+['Overall_percentage','Exam_Status']]

print("ERP Final Attendance Report")
```

```
print(erp_report)
```

```
erp_report.to_excel("Erp_final_attendence_report.xlsx",index=False)
```

Q12. "Write a program that:

- 1. Reads employee data from a Excel file**
- 2. Handles missing salary values**
- 3. Filters and sorts employees earning above 50,000**
- 4. Sort employees by salary in descending order**
- 5. display the top 5 high-earning employees**
- 6. Exports the result to Excel for analysis"**

Code: -

```
import pandas as pd

df= pd.read_excel('employee1.xlsx')

#print(df)

# Handle missing salary values by replacing them with 0
df['Salary']= df['Salary'].fillna(0)

#print(df)

#Filters and sorts employees earning above 50,000
high_salary=df[df['Salary'] > 50000]

#print(high_salary)

# Sort employees by salary in descending order
high_salary= high_salary.sort_values(by='Salary',ascending=False)

#print(high_salary)

high_salary.to_excel('high_salary_employees.xlsx',index=False)

#display the top 5 high-earning employees
print(high_salary.head())
```

Q13. "Write a program that:

- 1. Reads data from an Excel file.**
- 2. Displays only selected columns.**
- 3. Renames the columns appropriately."**

Code: -

```
import pandas as pd

#Reads data from an Excel file.
df = pd.read_excel('employee1.xlsx')

#Displays only selected columns.
#selected_columns=df[['EmployeeID','Name','Salary']]
selected_columns = df[['EmployeeID', 'Name', 'Salary']]

#print(selected_columns)

#Renames the columns appropriately.

rename=selected_columns.rename(columns={
    'EmployeeID':'ID',
    'Name': 'Employee_name',
    'Salary':'Employee_Salary'
})

print("\nRenamed Columns:")

print(rename)
```

Q14. """Write a Pandas program to:

- 1. Read data from an Excel file.**
- 2. Detect missing values in the dataset.**
- 3. Fill missing values using fillna()."""**

Code: -

```
import pandas as pd

df=pd.read_excel('employee1.xlsx')

#Detect missing values in the dataset.

print("Missing Values in Each Column:")

print(df.isnull())

print("\nTotal Missing Values in Each Column:")

print(df.isnull().sum())

#Fill missing values using fillna()

df['Salary']= df['Salary'].fillna(0)

print(df)
```

**Q15. '''Write a program that:
Reads employee data from a Excel file
Handles missing salary values
Filters and sorts employees earning above 50,000
Sort employees by salary in descending order
display above 50000 high-earning employees
Exports the result to Excel for analysis'''**

```
import pandas as pd
df= pd.read_excel('employee1.xlsx')
#print(df)

# Handle missing salary values by replacing them with 0
df['Salary']= df['Salary'].fillna(0)
#print(df)

#Filters and sorts employees earning above 50,000
high_salary=df[df['Salary'] > 50000]
print(high_salary)

# Sort employees by salary in descending order

high_salary= high_salary.sort_values(by='Salary',ascending=False)
#print(high_salary)
high_salary.to_excel('high_salary_employees.xlsx',index=False)

#display the top 5 high-earning employees
print(high_salary.head())
```

**Q16. '''Write a program that:
Reads data from an Excel file.
Displays only selected columns.
Renames the columns appropriately.
'''**

```
import pandas as pd
#Reads data from an Excel file.
df = pd.read_excel('employee1.xlsx')

#Displays only selected columns.

selected_columns = df[['EmployeeID', 'Name', 'Salary']]
#print(selected_columns)
```

```
#Renames the columns appropriately.
rename=selected_columns.rename(columns={
    'EmployeeID':'ID',
    'Name': 'Employee_name',
    'Salary':'Employee_Salary'
})
print("\nRenamed Columns:")
print(rename)
```

**Q17. """Write a Pandas program to:
Read data from an Excel file.
Detect missing values in the dataset.
Fill missing values using fillna()."""**

```
import pandas as pd
df=pd.read_excel('employee1.xlsx')
#Detect missing values in the dataset.
print("Missing Values in Each Column:")
print(df.isnull())

print("\nTotal Missing Values in Each Column:")
print(df.isnull().sum())
#Fill missing values using fillna()
df['Salary']= df['Salary'].fillna(0)
print(df)
```

Q12. An Excel file **Result.xlsx** contains student marks in five sheets: Unit Test, Prelim, Assignment, Attendance, and University Exam. Each sheet includes Rollno, Name, and subject-wise marks.

Write a Python program using **Pandas** to:

1. Read all five sheets from the Excel file.
2. Convert subject marks to numeric and replace missing values with **0**.
3. Scale marks as follows:
 - o Unit & Prelim → divide by 2
 - o University → convert 50 marks to 40
4. Calculate final total marks for each subject.
5. Save the result as **Final_Total_Marks_With_University.xlsx**.

Code: -

```
import pandas as pd
```

```
subjects = ['DS(TH)', 'DS(PR)', 'PHP(TH)', 'PHP(PR)', 'AWT', 'ADJ', 'SWT', 'PROJECT']
```

```
unit = pd.read_excel("Result.xlsx", sheet_name="Sheet1", header=1)
```

```
prelim = pd.read_excel("Result.xlsx", sheet_name="Sheet2", header=1)
```

```
assign = pd.read_excel("Result.xlsx", sheet_name="Sheet3", header=1)
```

```
att = pd.read_excel("Result.xlsx", sheet_name="Sheet4", header=1)
```

```
univ = pd.read_excel("Result.xlsx", sheet_name="Sheet5", header=1)
```

```
for df in [unit, prelim, assign, att, univ]:
```

```
    df[subjects] = df[subjects].apply(pd.to_numeric, errors='coerce').fillna(0)
```

```
unit[subjects] = (unit[subjects] / 2).round().astype(int)
```

```
prelim[subjects] = (prelim[subjects] / 2).round().astype(int)
```

```
univ[subjects] = (univ[subjects] * 40 / 50).round().astype(int)
```

```
total = unit[['Rollno', 'Name']].copy()
```

```
total[subjects] = (
```

```
    unit[subjects] +
```

```
    prelim[subjects] +
```

```
    assign[subjects] +
```

```
    att[subjects] +
```

```
    univ[subjects]
```

```
)
```

```
print(total)
```

```
total.to_excel("Final_Total_Marks_With_University.xlsx", index=False)
```

```
print("Saved as 'Final_Total_Marks_With_University.xlsx' successfully!")
```